

Written Assignment 3- Syntax Analysis

1. Eliminate the left recursion:

$$E \rightarrow ET+ \mid T$$

$$T \rightarrow TF^* \mid F$$

$$F \rightarrow E \mid i$$

2. Please analyse "ba(b)" by top-down method with backtracking, and tell us what are there weaknesses of top-down analysis, and why does LL(1) come? Then, judge whether the grammar is an LL(1) grammar by constructing the LL(1) parsing table.

$$M \rightarrow MaH \mid H$$

$$H \rightarrow b(M) \mid (M) \mid b$$

3. Please eliminate the left recursion and extract maximum common left factors (if there are) from the following context free grammar, and then decide the resulted grammar is whether a LL(1) grammar by **constructing the related LL(1) parsing table.**

$$S \rightarrow iCtS \mid iCtSeS \mid a$$

$$C \rightarrow C \text{ or } D \mid D$$

$$D \rightarrow D \text{ or } E \mid E$$

$$E \rightarrow (C) \mid b$$

4. Please show that the following operator grammar is whether an operator precedence grammar by **constructing the related parsing table.**

$$E \rightarrow E + F \mid F$$
$$F \rightarrow F * G \mid G$$
$$G \rightarrow @H \mid H$$
$$H \rightarrow (E) \mid i$$

5. **Construct a predictive table for the grammar:**

$$bexpr \rightarrow bexpr \text{ or } bterm \mid bterm$$
$$bterm \rightarrow bterm \text{ and } bfactor \mid bfactor$$
$$bfactor \rightarrow \text{not } bfactor \mid (bexpr) \mid \text{true} \mid \text{false}$$

6. **Generate operator-precedence relations for the following grammar:**

$$S \rightarrow aSbS \mid bSaS \mid \epsilon$$

7. Please **construct a LR(1) parsing table for the following ambiguous grammar with the additional conditions** that **else** dangles with the closest previous unmatched **if** , ; has the property of left associative.

$$S \rightarrow \text{if } S \text{ else } S \mid \text{if } S \mid S; S \mid a$$

8. Please **construct a LR(1) parsing table for the following ambiguous grammar with the additional conditions** that **or** has higher precedence than **do** , ; has higher precedence than **do** , ; has higher precedence than **or** , ; has the property of left associative.

$$S \rightarrow \text{do } S \text{ or } S \mid \text{do } S \mid S; S \mid a$$

Complete the homework and submit it before the next next class to the teaching assistant (Room 156, Building of Sch. Computer Sci&Eng).

1. Everyone should give their completed workbook to the teaching assist

ant before the designated time. There is a box at the door of the room specifically for collecting homework.

2. After the first submission of each assignment, it will be graded based on the quality of completion. Allow the next submission to modify errors from the previous assignment. The final score for each assignment will be given based on the final quality and number of revisions made.